

BM and Markov Processes

As we have seen before, BM has the following property:

$$W_t := B_{t+r} - B_r \quad t \geq 0$$

and

$$B_t \quad 0 \leq t \leq r$$

are independent (or processes)

In other words, since $B_{t+r} = W_t + B_r$,

$$P(B_{t+r} \in E \mid \mathcal{F}_r^B) = P(W_t + B_r \in E \mid \mathcal{F}_r^B)$$

We will make
this rigorous.

$$\left. \begin{array}{l} \text{We will make} \\ \text{this rigorous.} \end{array} \right\} = \underbrace{P(W_t + B_r \in E \mid B_r)}_{= P(W_t + u \in E) \Big|_{u=B_r}}$$

It is clear that we don't need in general to work with the natural filtration of BM but we can

Consider any admissible filtration, i.e., the following is true.

Theorem. Let $(B_t)_{t \geq 0}$ be a BM^d and $(F_t)_{t \geq 0}$ an admissible filtration. For every $s \geq 0$, the process $W_t := B_{t+s} - B_s$, $t \geq 0$, is also a BM^d and $(W_t)_{t \geq 0}$ is indep of F_s , i.e., $F_s := \sigma(W_t, t \leq s)$ is indep of F_s .

The point is that we can split any trajectories into two parts, which are independent:

take $s, t \geq 0$

$$B_{t+s} = \underbrace{B_{t+s} - B_s}_{W_t} + B_s$$

$$= \underbrace{W_t}_{\text{independent}} + \underbrace{B_s}_y = W_t + y \Big|_{y=B_s}$$

Note that $W_t + y$ is a BM started at y .

The notation

$$P^n(B_t \in A_1, \dots, B_{t_n} \in A_n) := P(x+B_t \in A_1, \dots, x+B_{t_n} \in A_n)$$

is very common for Markov processes.

The measure corresponding to P^x on the path space,
i.e., the Wiener measure μ^x is supported on the
continuous functions $w: [0, \infty) \rightarrow \mathbb{R}^d$ s.t. $w(0) = x$.
It follows also that

$$x \mapsto \mathbb{E}^x u(B_t)$$

is measurable for every $u \in B_b(\mathbb{R}^d)$

bounded Borel measurable
functions $u: \mathbb{R}^d \rightarrow \mathbb{R}$.

this section needs a sketched proof:

Note that

$$\mathbb{E}^x u(B(t)) = \mathbb{E} u(x + B(t))$$

$$= \int u(x+y) P(B(t) \in dy)$$

and this last representation is clearly measurable (Fubini).

Furthermore $x \mapsto \mathbb{E}^x u(B_t)$ is also bounded, i.e.,

$$x \mapsto \mathbb{E}^x u(B_t) \text{ is in } B_b(\mathbb{R}^d).$$

The last section can be generalized to any r.v. $Z \in \mathcal{F}_\infty^B$
bounded.

The next technical result will be crucial to make things progress.

Lemma. Let $X: \Omega \mapsto \mathcal{D}$ s.t. $X^{-1}: \mathcal{D} \mapsto \mathcal{A}$
 and $Y: \Omega \mapsto \mathcal{E}$ s.t. $Y^{-1}: \mathcal{E} \mapsto \mathcal{A}$.
 be two random variables. Suppose that $\mathcal{X}, \mathcal{Y} \subset \mathcal{A}$
 are σ -Algebras s.t. $X^{-1}: \mathcal{D} \mapsto \mathcal{X}$ and
 $Y^{-1}: \mathcal{E} \mapsto \mathcal{Y}$, $\left| \begin{array}{l} \sigma(X) \subset \mathcal{X} \\ \sigma(Y) \subset \mathcal{Y} \end{array} \right.$
 and further $\mathcal{X} \perp \mathcal{Y}$.

Then

$$\mathbb{E}[\phi(X, Y) | \mathcal{X}] = \mathbb{E}[\phi(X, Y) |_{\Omega=X}]$$

$$= \mathbb{E}[\phi(X, Y) | \mathcal{X}]$$

for every bounded functions ϕ s.t.

$$\phi^{-1}: \mathcal{B}(\mathbb{R}) \mapsto \mathcal{D} \times \mathcal{E}$$

Proof. Suppose first that $\phi(X, Y) = \phi_1(X) \phi_2(Y)$.
 then

$$\begin{aligned} \mathbb{E}[\phi(X, Y) | \mathcal{X}] &= \mathbb{E}[\phi_1(X) \phi_2(Y) | \mathcal{X}] \\ &= \phi_1(X) \mathbb{E}[\phi_2(Y) | \mathcal{X}] \end{aligned}$$

$\phi_1(X) \in \mathcal{X}$

$$\begin{aligned}
 & X \perp Y \quad Y \in \mathcal{G} \\
 &= \phi_1(X) \otimes \phi_2(Y) \\
 &= \left[E \phi(X, Y) \right]_{X=X} \longrightarrow \left[E \phi_1(X) \phi_2(Y) \right]_{X=X}
 \end{aligned}$$

By definition of conditional expectation this means that,
for every $F \in \mathcal{X}$, one has

$$E[1_F \phi(X, Y)] = E[1_F [E\phi(X, Y)]_{X=X}]$$

whenever $\phi(X, Y) = \phi_1(X) \phi_2(Y)$. *

In other words, letting $\phi(X, Y) = 1_A(X) 1_B(Y)$ for $A \in \mathcal{D}$ and $B \in \mathcal{E}$, the equality * reads

$$\int_F 1_{A \times B}(X, Y) dP = \int_F \left[E 1_{A \times B}(X, Y) \right]_{X=X} dP$$

for every $F \in \mathcal{X}$. Then the class $\{A \times B : A \in \mathcal{D}, B \in \mathcal{E}\}$
and the mapping

$$\Sigma' \ni A \times B \mapsto \int_F 1_{A \times B}(X, Y) dP$$

$$\Sigma' \ni A \times B \mapsto \int_F \left[E 1_{A \times B}(X, Y) \right]_{X=X} dP$$

are countably additive maps (prove this). Then

then the mappings can be extended to a (unique) measure on $\sigma(Z')$. It follows that

$$\int_F 1_{\sigma}(x, y) dP = \int_F \left[E 1_{\sigma}(x, y) \right]_{n=x} dP \quad \square$$

for every $\sigma \in \tilde{\Sigma}^1$ and $F \in \mathcal{X}$.
 By a classical standard machine argument we can extend the equality $\square$ to any function $\phi \in B_b(\mathbb{D} \times \mathbb{E})$, to get

$$E 1_F \phi(x, y) = E \left[1_F \left[E \phi(x, y) \right]_{n=x} \right]$$

which is equivalent to

$$E \left[\phi(x, y) \mid \mathcal{X} \right] = E \left[\left[E \phi(x, y) \right]_{n=x} \mid \mathcal{X} \right]$$

$$= \left[E \phi(x, y) \right]_{n=x}$$

↓
which is the first equality

Now take conditional expectation on both sides of the previous equality to get

$$\mathbb{E}[\mathbb{E}[\phi(x, y) | \mathcal{X}] | X] = \mathbb{E}[\underbrace{[\mathbb{E}\phi(x, y)]_{n=X}}_{\text{measurability}} | X]$$

$\downarrow$ tower

$$\mathbb{E}[\phi(x, y) | X] = [\mathbb{E}\phi(x, y)]_{n=X}$$

$$= \mathbb{E}[\phi(x, y) | \mathcal{X}]$$

$\hookrightarrow$ by the 1st equality.

We are ready to state the Markov property formally.

Theorem. Let $(B_t)_{t \geq 0}$ and $(\mathcal{F}_t)_{t \geq 0}$ be a BM and an admissible filtration. Then

$$\mathbb{E}[u(B_{t+r}) | \mathcal{F}_r] = \mathbb{E}^{B_r}[u(B_t)]$$

$$\left(= [\mathbb{E} u(B_t + r)]_{r=B_r} \right)$$

for any $u \in \mathcal{B}_b(\mathbb{R}^d)$.

Proof. Write $u(B_{t+r}) = u(\underbrace{B_r}_{=: X} + \underbrace{B_{t+r} - B_r}_{=: Y})$.

Then $X \perp Y$ $X \in \mathcal{F}_r$ $Y \perp \mathcal{F}_r$ and thus

$$E[u(B_{t+r}) | \mathcal{F}_r] = E[u(X+Y) | \mathcal{F}_r]$$

use previous lemma with $\phi(x, y) = u(x+y)$

$$= [E u(x+y)]_{x=X}$$

$$= [E u(x + B_{t+r} - B_r)]_{x=B_r}$$

Adaptation
in variables

$$= [E u(x + B_t)]_{x=B_r}$$

$$= E^{B_r} u(B_t).$$

Now we extend the previous result to any functional of Brownian trajectories. For this we need:

Lemma. Under the assumptions of the lemma above, suppose further that

$$\psi: \Delta \times \Omega \mapsto \mathbb{R} \quad \text{is a t.}$$

$$\psi^{-1}: \mathcal{B}(\mathbb{R}) \mapsto \mathcal{Q} \otimes \mathcal{G} \quad \text{ad is bounded.}$$

Then $E[\psi(X(\cdot), \cdot) | \mathcal{X}]$

$$= [E\psi(u, \cdot)]_{u=X}$$

$$= E[\psi(X(\cdot), \cdot) | \mathcal{X}]$$

[Proof. Vary u in the previous case since the dependence of $\psi(X(\cdot), u)$ on u is s.t. it is "independent" among the two conditions.]

Use now this lemma to prove that

[Theorem. Let $(B_t)_{t \geq 0}$ and $(\mathcal{F}_t)_{t \geq 0}$ be BM and adn. filtration. then

$$E[\psi(B_{\cdot+r}) | \mathcal{F}_r] = E^{\mathbb{B}_r}[\psi(B_{\cdot})]$$

for all $\psi: C[0, \infty) \rightarrow \mathbb{R}$ s.t. $\psi^{-1}: \mathbb{B} \rightarrow \mathbb{B}(c)$ bounded.

Proof. Use previous lemma or follow:

$$\psi(B_{\cdot+r}) = \psi(B_{\cdot+r} - B_r + \underbrace{B_r}_X)$$

Then $X = \mathcal{F}_r$ $Y = \mathcal{F}_{t \geq 0}^{B_{t+r} - B_r}$ and thus

$$\begin{aligned} E[\psi(B_{\cdot+r}) | \mathcal{F}_r] &= E[\psi(B_{\cdot+r} - B_r + B_r) | \mathcal{F}_r] \\ &= E[\psi(B_{\cdot} + r)]_{r=B_r} \\ &= E^{B_r}[\psi(B_{\cdot})] \end{aligned}$$

Note that we used everywhere the same notation

$B_{\cdot} + r$ $B_{\cdot+r}$ to denote

↓
space translation
of a function

↘
time translation
of a function



Note that previous Theorem does not need to be applied to processes with stationary increments. In general, any $\mathcal{F}_t$ -adapted random process is said to be a Markov process if

$$\mathbb{P}[u(X_{t+r}) | \mathcal{F}_r] = \mathbb{P}[u(X_{t+r}) | X_r]$$

for every $t \geq 0$ and $r \geq 0$ and $u \in \mathcal{B}_b(\mathbb{R}^d)$.

Define

$$J_{\underline{u}, \underline{r}, \underline{t+r}}(\underline{u}) := \mathbb{E}[\underline{u}(X_{\underline{t+r}}) | X_{\underline{r}} = \underline{u}]$$

and define

$$\mathbb{P}^{r,u} u(X_{t+r}) := J_{u,r,t+r}(u)$$

If $J_{u,r,t+r}(u)$ depends only on $(t+r) - r = t$, i.e., the difference, then we call the process homogeneous and the Markov property is written as

$$\mathbb{P}[u(X_{t+r}) | \mathcal{F}_r] = \mathbb{E}^{X_r} u(X_t)$$

It is clear that, for homogeneous process

this equality implies the more general

$$\mathbb{P}[u(X_{t+r}) | \mathcal{F}_r] = \mathbb{P}[u(X_{t+r}) | \mathcal{X}_r].$$

Further for Markov processes one has

$$\mathbb{P}[\psi(X_{\cdot+r}) | \mathcal{F}_r] = \mathbb{P}[\psi(X_{\cdot+r}) | \mathcal{X}_r]$$

$$(\text{homogeneity}) \quad = \mathbb{E}^{X_r}[\psi(X_{\cdot})]$$

for any bounded measurable functionals
 $\psi: C[0, \infty) \rightarrow \mathbb{R}$ of the trajectories.

NB: Although this last relationship looks more general it is possible to obtain it from the definition.

Remark. Recall the Markov property

$$\mathbb{P}[u(X_{t+r}) | \mathcal{F}_r] = \mathbb{E}^{X_r} u(X_t)$$

by specializing the dependence on " ω " or

$$\mathbb{E}[m(X_{t+r}) | \mathcal{F}_r](\omega)$$

$$= \int_{\mathbb{R}^d} m(X_t(\omega)) dP^{X_r(\omega)}$$

$$= \int_{\mathbb{R}^d} m(y) P^{X_r(\omega)}(X_t \in dy) \quad (*)$$

Now take three different times $0 \leq r \leq t \leq n$
and $\phi, \psi, \chi: \mathbb{R}^d \mapsto \mathbb{R}$.

$$\mathbb{E}[\phi(B_n) \psi(B_t) \chi(B_n)] = \mathbb{E}[\mathbb{E}[\phi(B_n) \psi(B_t) \chi(B_n) | \mathcal{F}_t]]$$

$$= \mathbb{E}[\phi(B_n) \psi(B_t) \mathbb{E}[\chi(B_n) | \mathcal{F}_t]]$$

Markov prop.

$$= \mathbb{E}[\phi(B_n) \psi(B_t) E^{B_t} \chi(B_{n-t})]$$

$$= \mathbb{E}[\phi(B_n) \psi(B_t) E^{B_t} \chi(B_{n-t}) | \mathcal{F}_r]$$

$$= \mathbb{E}[\phi(B_n) E[\psi(B_t) E^{B_t} \chi(B_{n-t}) | \mathcal{F}_r]]$$

Markov property

$$= \mathbb{E}[\phi(B_n) E^{B_r} \psi(B_{t-r}) E^{B_{t-r}} \chi(B_{n-t})]$$

use $\otimes$ repeatedly

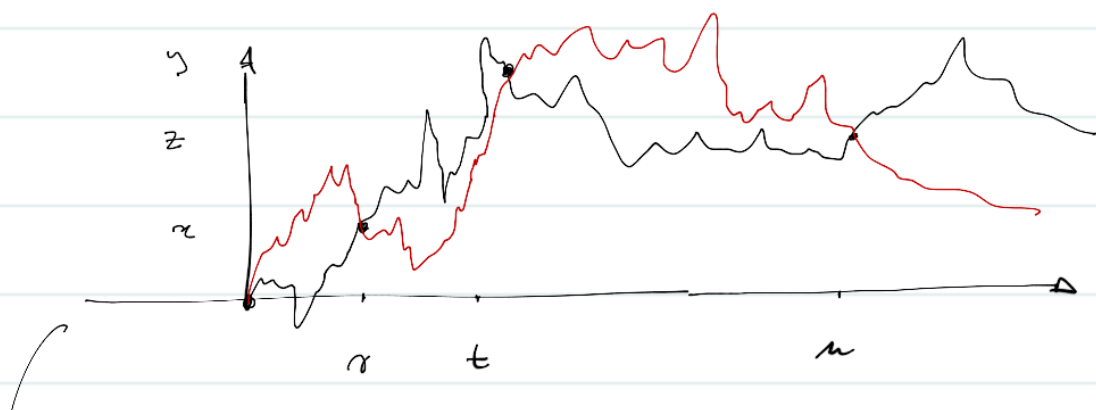
$$= \left\{ \int \phi(x) \left[\int \psi(y) \left(\int \chi(z) P^y(B_{u-t} \in dz) \right) P^x(B_{t-r} \in dy) \right] P(B_r \in dx) \right\}$$

The above equality reads

$$\begin{aligned} & \iiint \phi(x) \psi(y) \chi(z) P(B_0 \in dx, B_t \in dy, B_u \in dz) \\ &= \iiint \phi(x) \psi(y) \chi(z) P^y(B_{u-t} \in dz) P^x(B_{t-r} \in dy) P(B_0 \in dx) \end{aligned}$$

and this essentially means that the joint distribution (B_0, B_t, B_u) is

$$P(B_0 \in dx, B_t \in dy, B_u \in dz) = P^y(B_{u-t} \in dz) P^x(B_{t-r} \in dy) P(B_0 \in dx)$$



Prop. of these projectors:

$$0 \longrightarrow \mathcal{U}$$

$$\mathcal{U} \xrightarrow{t \rightarrow 0} \mathcal{V}$$

$$\mathcal{V} \xrightarrow{u \rightarrow t} \mathcal{W}$$

With this in mind it is possible to prove:

Theorem. Let (X_t) be a (homogeneous) Markov process, let $t_0 = 0 < t_1 < \dots < t_n$. Then

$$P(X_{t_1} \in du_1, \dots, X_{t_n} \in du_n) = \frac{1}{u} \prod_{j=1}^n P^{u_{j-1}}(X_{t_j - t_{j-1}} \in du_j).$$

induction maximal!